

Gaussian distribution is arguably the most important single distribution (due to CLT) and it has a lot of interesting properties.

### bivariate Gaussian distributions

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Showing it is a proper distribution

$$\int_{\mathbb{R}} f(x; \mu, \sigma) dx \cdot \int_{\mathbb{R}} f(y; \mu, \sigma) dy$$

instead of solving  $\int_{\mathbb{R}} f(x; \mu, \sigma) dx$  directly, we use this multiplication trick here to simplify the problem

$$= \frac{1}{2\pi\sigma^2} \int_{\mathbb{R}} \int_{\mathbb{R}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} - \frac{(y-\mu)^2}{2\sigma^2}\right) dx dy$$

$$= \frac{1}{2\pi\sigma^2} \int_{\mathbb{R}} \int_{\mathbb{R}} \exp\left(-\frac{1}{2\sigma^2} ((x-\mu)^2 + (y-\mu)^2)\right) dx dy$$

Let  $(x-\mu) = r \cos \theta$   
 $(y-\mu) = r \sin \theta \Rightarrow J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$

$$\Rightarrow |J| = r \cos^2 \theta + r \sin^2 \theta = r$$

Hence we have

$$\begin{aligned} & \int_{\mathbb{R}} f(x; \mu, \sigma) dx \int_{\mathbb{R}} f(y; \mu, \sigma) dy \\ &= \frac{1}{2\pi\sigma^2} \int_0^{2\pi} \int_0^\infty \exp\left(-\frac{1}{2\sigma^2} r^2\right) r dr d\theta \\ &= \frac{1}{2\pi\sigma^2} \int_0^{2\pi} \int_0^\infty \frac{1}{2} \exp\left(-\frac{1}{2\sigma^2} r^2\right) dr^2 d\theta \\ &= \frac{1}{4\pi\sigma^2} \int_0^{2\pi} \left(-2\sigma^2 \exp\left(-\frac{1}{2\sigma^2} r^2\right)\right) \Big|_{r=0}^\infty d\theta \\ &= \frac{1}{4\pi\sigma^2} \int_0^{2\pi} 2\sigma^2 d\theta \\ &= \frac{1}{4\pi\sigma^2} \cdot 4\pi\sigma^2 = 1 \end{aligned}$$

derive its moment generating function

$$\begin{aligned}
 \bar{E}(e^{xt}) &= \int_{-\infty}^{+\infty} e^{xt} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{xt - \frac{1}{2\sigma^2}(x^2 + \mu^2 - 2\mu x)} dx \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{1}{2\sigma^2}(x^2 - 2\mu x - 2\sigma^2 x + \mu^2)} dx \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{1}{2\sigma^2}((x - (\mu + \sigma^2 t))^2 - \sigma^2 t^2 - 2\mu\sigma^2 t)} dx \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{\frac{1}{2\sigma^2}\mu^2 + \mu t} \cdot e^{-\frac{1}{2\sigma^2}(x - (\mu + \sigma^2 t))^2} dx \\
 &= e^{\frac{1}{2}\sigma^2 t^2 + \mu t} \cdot \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{1}{2\sigma^2}(x - (\mu + \sigma^2 t))^2} dx \\
 &= e^{\frac{1}{2}\sigma^2 t^2 + \mu t}
 \end{aligned}$$

if  $\mu = 0$ ,  $\sigma = 1$ , we have

$$\bar{E}(e^{xt}) = e^{\frac{1}{2}t^2}$$

$$\Rightarrow E(x) = (e^{\frac{1}{2}t^2}) \Big|_{t=0} = 0$$

$$\bar{E}(x^2) = (e^{\frac{1}{2}t^2} + t^2 e^{\frac{1}{2}t^2}) \Big|_{t=0} = 1$$

$$\bar{E}(x^3) = (t e^{\frac{1}{2}t^2} + 2t e^{\frac{1}{2}t^2} + t^3 e^{\frac{1}{2}t^2}) \Big|_{t=0} = 0$$

$$\bar{E}(x^4) = (e^{\frac{1}{2}t^2} + t^2 e^{\frac{1}{2}t^2} + 2t e^{\frac{1}{2}t^2} + 2t^3 e^{\frac{1}{2}t^2} + t^4 e^{\frac{1}{2}t^2}) \Big|_{t=0} = 3$$

# Multivariate Gaussian distributions

$$f(x; \mu, \Sigma) = (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu))$$

proving it is a proper distribution:

$$\begin{aligned} I &= \int_{\mathbb{R}^n} \exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)) dx \\ &= \int_{\mathbb{R}^n} \exp(-\frac{1}{2}y^T \Sigma^{-1}y) dy \end{aligned}$$

if  $z \sim N(0, I)$   
then  $\Sigma^{\frac{1}{2}} z + \mu \sim N(\mu, \Sigma)$   
on the other hand, if  $z \sim N(\mu, \Sigma)$   
 $\Sigma^{-\frac{1}{2}}(z-\mu) \sim N(0, I)$

Since  $\Sigma$  is a valid covariance matrix, we must have

$$\begin{aligned} \Sigma &= S \Lambda S^T, \text{ where } S \text{ is orthonormal matrix, } \Lambda \text{ is a diagonal matrix with } \Lambda_{i,i} = \lambda_i > 0 \\ \Rightarrow \Sigma^{-1} &= S \Lambda^{-1} S^T \\ \Rightarrow y^T \Sigma^{-1} y &= y^T S \Lambda^{-1} S^T y \end{aligned}$$

$$= z^T \Lambda^{-1} z, \text{ where } z = S^T y$$

Note  $\frac{\partial y}{\partial z} = \frac{\partial(Sz)}{\partial z} = S \Rightarrow |\det(S)| = 1$

because  $S$  is orthogonal matrix, we have

$$SS^T = I$$

$$\Rightarrow |S| |S^T| = 1$$

$$\Rightarrow |\det(S)| = \pm 1$$

So we have

$$I = \int_{\mathbb{R}^n} \exp(-\frac{1}{2}z^T \Lambda^{-1}z) dz$$

$$= \int_{\mathbb{R}^n} \exp(-\frac{1}{2} \sum_{i=1}^n \frac{1}{\lambda_i} z_i^2) dz_1, \dots, dz_n$$

$$= \prod_{i=1}^n \sqrt{2\pi \lambda_i}$$

$$= (2\pi)^{\frac{n}{2}} |\Sigma|^{-\frac{1}{2}}$$

$$\begin{aligned} |\Sigma| &= |S| |\Lambda| |S^T| \\ &= \prod_{i=1}^n \lambda_i \end{aligned}$$

## mean and variance of the multivariate Gaussian

$$\begin{aligned} \bar{E}(x) &= \int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} x \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right) dx \\ &= \int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} (y+\mu) \exp\left(-\frac{1}{2}y^T \Sigma^{-1}y\right) dy \\ &= \mu + \underbrace{\int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} y \exp\left(-\frac{1}{2}y^T \Sigma^{-1}y\right) dy}_{= 0 \text{ due to symmetry}} \\ &= \mu \end{aligned}$$

$$\text{Cov}(X) = \bar{E}\left((X-\mu)(X-\mu)^T\right)$$

$$\begin{aligned} &= \int_{\mathbb{R}^n} (x-\mu)(x-\mu)^T (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right) dx \\ &= \int_{\mathbb{R}^n} yy^T (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}y^T \Sigma^{-1}y\right) dy \end{aligned}$$

Let  $z = \Sigma^{-\frac{1}{2}}y$ ,  $\Rightarrow y = \Sigma^{\frac{1}{2}}z$

$$\begin{aligned} \Rightarrow &= \int_{\mathbb{R}^n} \Sigma^{\frac{1}{2}} z z^T \Sigma^{\frac{1}{2}} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}z^T z\right) |\det(\Sigma^{\frac{1}{2}})| dz \\ &= \Sigma^{\frac{1}{2}} \underbrace{\int_{\mathbb{R}^n} z z^T (2\pi)^{-\frac{n}{2}} \exp\left(-\frac{1}{2}z^T z\right) dz}_{\text{all independent now}} \Sigma^{\frac{1}{2}} \\ &= \Sigma \end{aligned}$$



## Conditional distribution

If  $(X, Y) \sim \mathcal{N} \left( \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix} \right)$ , then  $Y|X \sim ?$

$$f_{X,Y}(x, y; \mu, \Sigma) = (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp \left( -\frac{1}{2} \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix}^T \Sigma^{-1} \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix} \right)$$

Let  $\Lambda = \Sigma^{-1}$

$$= (SAS^T)^{-1}$$

$$= S \Lambda^{-1} S^T \rightarrow \text{it is positive-definite as well}$$

$$= \begin{pmatrix} \Lambda_{xx} & \Lambda_{xy} \\ \Lambda_{yx} & \Lambda_{yy} \end{pmatrix}, \text{ where } \Lambda_{xx} \text{ and } \Lambda_{yy} \text{ are symmetric}$$

$$\Rightarrow -\frac{1}{2} \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix}^T \Sigma^{-1} \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix} = -\frac{1}{2} \left( \begin{pmatrix} x - \mu_x \end{pmatrix}^T \Lambda_{xx} \begin{pmatrix} x - \mu_x \end{pmatrix} + \begin{pmatrix} y - \mu_y \end{pmatrix}^T \Lambda_{yy} \begin{pmatrix} y - \mu_y \end{pmatrix} + \begin{pmatrix} x - \mu_x \end{pmatrix}^T \Lambda_{xy} \begin{pmatrix} y - \mu_y \end{pmatrix} + \begin{pmatrix} y - \mu_y \end{pmatrix}^T \Lambda_{yx} \begin{pmatrix} x - \mu_x \end{pmatrix} \right)$$

No that the conditional function  $f_{Y|X}(y|x; \mu, \Sigma) = \frac{f_{X,Y}(x, y; \mu, \Sigma)}{f_X(x; \mu, \Sigma)}$  is a function of  $y$

$$\text{and } -\frac{1}{2} \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix}^T \Sigma^{-1} \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix} = -\frac{1}{2} \left( y^T \Lambda_{yy} y - y^T \Lambda_{yy} \mu_y - \mu_y^T \Lambda_{yy} y + y^T \Lambda_{yx} (x - \mu_x) + (x - \mu_x)^T \Lambda_{xy} y + \text{const} \right)$$

$$= -\frac{1}{2} \left( y^T \Lambda_{yy} y + y^T (2 \Lambda_{yx} (x - \mu_x) - \Lambda_{yy} \mu_y) \right) + \text{const}$$

$$= -\frac{1}{2} \left( y^T \Lambda_{yy} y + 2 y^T \Lambda_{yy} (\Lambda_{yy}^{-1} \Lambda_{yx} (x - \mu_x) - \mu_y) \right) + \text{const}$$

Note that any multivariate Gaussian distribution has the form

$$(2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right\}$$

$$\rightarrow -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)$$

$$= -\frac{1}{2} (x^T \Sigma^{-1} x - x^T \Sigma^{-1} \mu - \mu^T \Sigma^{-1} x + \mu^T \Sigma^{-1} \mu)$$

$$= -\frac{1}{2} (x^T \Sigma^{-1} x - 2 x^T (\Sigma^{-1} \mu)) + \text{constant}$$

$\downarrow$  quadratic term  $\quad \downarrow$  linear term

Therefore, we must have the covariance of the conditional function

$$\text{Cov}(Y|X) = \Lambda_{yy}^{-1}$$

Symmetric  
( $\Leftrightarrow$ )

$$\text{Cov}(X|Y) = \Lambda_{xx}^{-1}$$

$$E(Y|X) = \Lambda_{yy}^{-1} \Lambda_{yx} (x - \mu_x) - \mu_y$$

$$E(X|Y) = \Lambda_{xx}^{-1} \Lambda_{xy} (y - \mu_y) - \mu_x$$

Now, the problem is to express the precision matrix

$$\Sigma^{-1} = \Lambda = \begin{pmatrix} \Lambda_{xx} & \Lambda_{xy} \\ \Lambda_{yx} & \Lambda_{yy} \end{pmatrix} \quad \text{using } \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

To do this, we apply Schur Complement based on Gaussian elimination

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix} = \begin{pmatrix} A - BD^{-1}C & B \\ 0 & D \end{pmatrix}$$

$$\begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} A - BD^{-1}C & B \\ 0 & D \end{pmatrix} = \begin{pmatrix} A - BD^{-1}C & 0 \\ 0 & D \end{pmatrix}$$

$$\text{So we have } \begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix} = \begin{pmatrix} A - BD^{-1}C & 0 \\ 0 & D \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & BD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} A - BD^{-1}C & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} I & 0 \\ D^{-1}C & I \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix} \begin{pmatrix} (A - BD^{-1}C)^{-1} & 0 \\ 0 & D^{-1} \end{pmatrix} \begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}$$

$$= \begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix} \begin{pmatrix} M & 0 \\ 0 & D^{-1} \end{pmatrix} \begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}$$

$$= \begin{pmatrix} M & 0 \\ -D^{-1}CM & D^{-1} \end{pmatrix} \begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}$$

$$= \begin{pmatrix} M & -MBD^{-1} \\ -D^{-1}CM & D^{-1}CMBD^{-1} + D^{-1} \end{pmatrix}$$

$$\text{where } M = (A - BD^{-1}C)^{-1}$$